Understanding Players

Raw player data to six actionable segments

For technical and non-technical audiences · product, marketing & business view

Betty · 100,000 player records · snapshot 2025-11-13

Executive summary: what we found

TL;DR

- 100,000 player records were explored; 59,922 had full, meaningful activity data and were profiled in detail.
- They split into 6 clearly different segments – customers who deserve different treatment.
- Whales (18% of players) generate ~90% of gross gaming revenue – protecting them is priority #1.
- Some players cost money: Net-Loss Gamers (~14%) produce negative GGR, and bonus abuse concentrates in them and in Dormant accounts.
- The smallest spenders are proportionally the most profitable (best GGR per deposited dollar).

  i.e. relative to how little they put in, the operator keeps the highest share of their money.
- Recommendation – protect the whales, restrict the loss-makers, grow the healthy core, convert the slow ones.

The question – and the path to an answer

Roadmap

The business question

Are all players the same? Do different players behave differently – and if so, should we talk to them, reward them and manage risk for them differently?

The task: let the data answer.
Approach: Group players who behave alike, then translate each group into actions – for product, for marketing, and for the business.

The data science cycle, in plain language

- 1. Understand the data – what do we have, what is missing?
- 2. Build meaningful measurements – features that capture real behaviour.
- 3. Shape the numbers – make huge and tiny values comparable fairly.
- 4. Group similar players – clustering (K-Means and friends).
- 5. Check the groups are solid – metrics + reproducibility.
- 6. Turn groups into actions – segment profiles, differentiation, recommendations.

Step 1 · Understand the data

The Data

- 100,000 player accounts × 77 attributes: account details, marketing/UTM, KYC & compliance, responsible-gambling flags, and money activity.
- Snapshot date 2025-11-13 – every “lifetime” number is measured up to this point.
- Most players are in Canada; core activity fields (deposits, withdrawals, bets, wins, logins) are reliable for ~60% of profiles (missing data).
- Plain-language view: one giant loyalty-card ledger – sign-up, marketing source, risk flags, and who spent/cashed out what.

Technical snapshot

- Source: Data Science Task Data.xlsx - Result 1.csv
- REFERENCE_DATE = max(loaded_at) = 2025-11-13
- Money & frequency fields heavily right-skewed (Fisher skew 7–50)
- Several alias / near-duplicate columns (e.g. login_rate vs login_frequency) collapsed in EDA
- marketing_network column is 100% empty → excluded
- Nullable flags: unrecorded ≠ False (NaN kept, never imputed)

Step 1 (deep dive) · Exploring the data (EDA)

The Data

- Goal: know the data before modelling – quality, gaps, and how each number behaves.
- Big finding: a handful of players dominate spending – think “0.1% of whales, a long tail…”
- Missingness matters: many risk/abuse booleans were simply never recorded – that is “unknown”, not “no”.
- Demographics are weak differentiators (almost everyone Canadian, no meaningful age/income split).
- Key Lesson: behaviour (who spends, who logs in, who withdraws) drives segmentation, not demographics.

Technical snapshot

- Univariate + completeness audits per column
- GAIN from explicit data-type audit: dates normalized, aliases merged
- House-edge sanity check: mean GGR / mean bets ≈ 5.2% (slot economics)
- 0.03% implausible birth years (e.g. 0196) → age clipped to [18,100]
- 43 derived features documented (feature_dictionary.csv)
- Enumeration of KYC, RG, bonus-abuse flag coverage

Step 2 · Build meaningful measurements

Features

Raw columns were turned into a compact set of measurements, each with a real business meaning. Ten were chosen for clustering – covering life cycle, engagement and value.

Group	Feature (what it measures)
Life cycle	tenure_days – how long the account has existed
Life cycle	time_to_first_deposit – how fast a player first funds
Engagement	total_logins – overall activity level
Engagement	login_rate – sustained habit (logins per month)
Value	total_deposits / total_withdrawals – money in / money out
Value	net_deposits – net money position (signed)
Behaviour	turnover-to-deposit ratio – betting velocity
Behaviour	withdrawal_ratio – how much is cashed out vs staked
Behaviour	bets_per_login – depth of each session

- 43 derived features in total (features/feature_dictionary.csv).
- Only 10 kept for modelling – the rest are aliases, too-missing, or not actionable.
- ratios are not time-normalised; rates are per month – both documented.

Step 3 · Shape the numbers so the model stays fair

Distribution · Scaling

- The problem: a £100,000 depositor makes a £100 depositor look identical – raw numbers are unfairly heavy-tailed.
- Compress the tails: log-transform money & activity so a “big” and a “small” player are compared on a fair scale (safe for zeros, invertible).
- Standardize: after compression, rescale every measurement so none dominates the grouping by sheer unit size.
- Filter to complete records: 59,922 players with every measurement present were used; the other 40,078 lack too much data to judge.

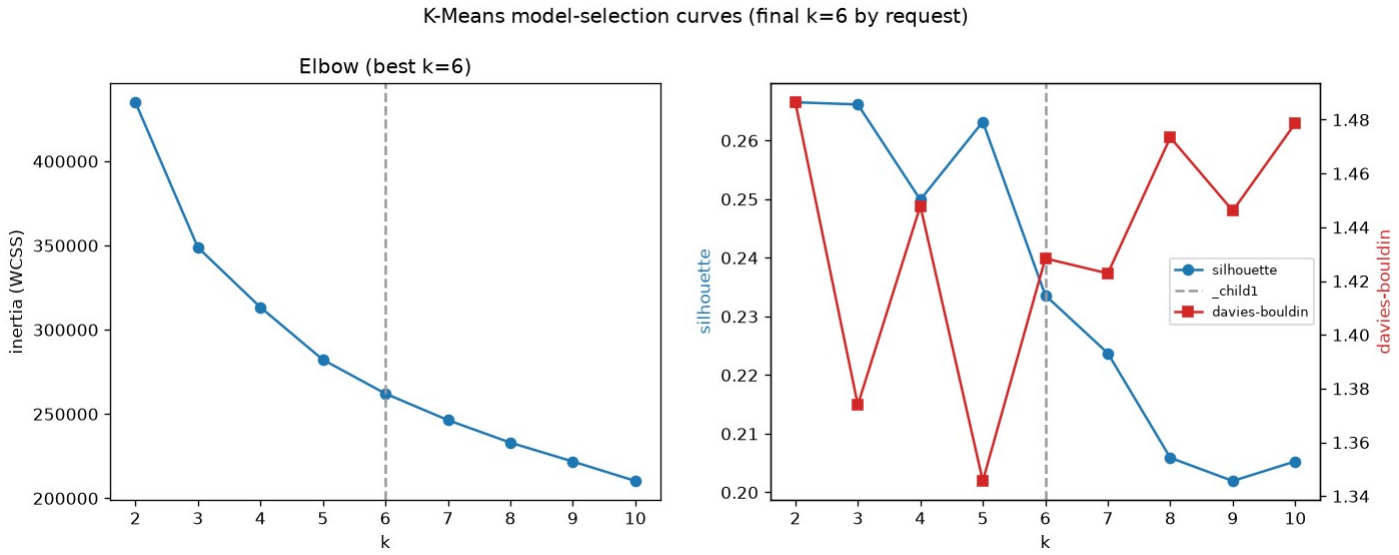
Technical: transforms applied

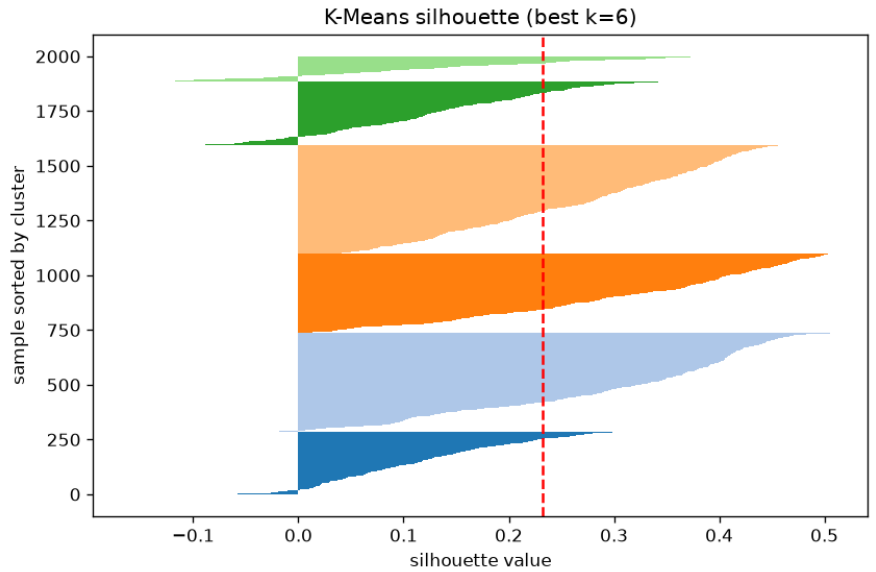
- log1p(x)=ln(1+x) for non-negative; slog1p=sign(x)·ln(1+|x|) when negatives exist
- Rule: transform when Fisher skew > 3; keep otherwise
- e.g. deposits 24.1 → 0.6 · turnover ratio 49.5 → 0.3 · withdrawals 26.6 → 1.0
- z-score (StandardScaler): mean 0, std 1 on the same 59,922 matrix
- All transforms reversible → clusters can be read back in £/days

Step 4 · Build the grouping – K-Means

Modelling

Plain language: K-Means “finds 6 natural groups” so that players inside a group are alike and groups differ from each other – like sorting customers by “who behaves like whom”.





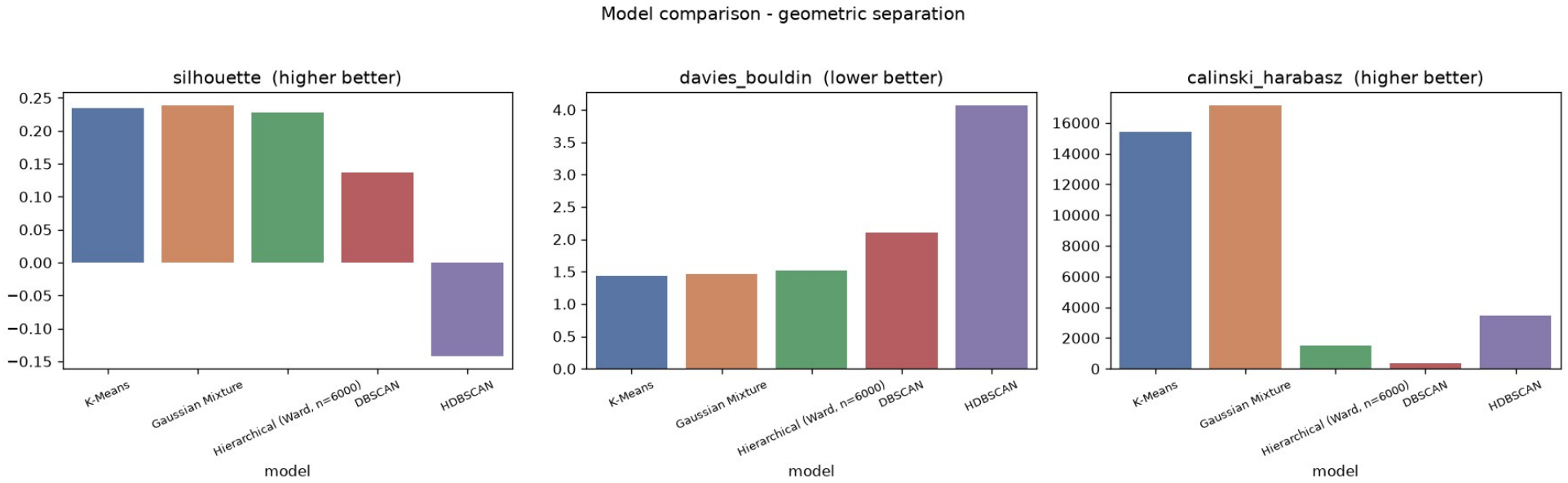
- Tried k = 2…10. k = 2, 3 and 5 score almost identically (silhouette 0.267 / 0.266 / 0.263).
- 6 groups – six genuinely actionable segments (silhouette 0.234).
- Technical: inertia + silhouette + Davies-Bouldin curves evaluated; n_init=10, deterministic seed; metrics in kmeans_metrics.csv.

Step 5 · We compared five grouping methods – then checked they hold up

Evaluation

- Plain language: more sophisticated algorithms were tried (soft groups, tree structures, density-based). K-Means wins for practicality and, crucially, it is the most stable – rerunning it gives (almost) the same groups every time.
- Stability = trust: if a model gives a different answer every run, you cannot build business rules on it. K-Means 0.996 vs GMM 0.804, Hierarchical 0.579.

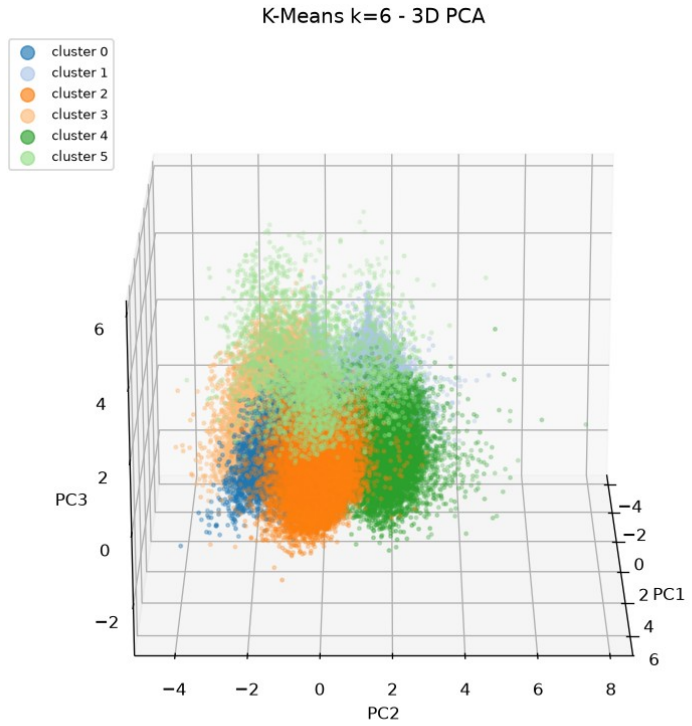
Model (fitted)	Groups	Noise pts	Silhouette	Davies-Bouldin	Stability (ARI)
K-Means, k=6	6	0	0.234	1.428	0.996
Gaussian Mixture, k=3	3	0	0.239	1.462	0.804
Hierarchical (Ward), k=4	4	0	0.228	1.513	0.579
DBSCAN, eps=1.0, min-samples=20	3	2,375	0.137	2.108	0.855
HDBSCAN, min-cluster=500	7	31,438	-0.142	4.062	0.640



The K-Means result: six player segments

6 Groups

#	Segment name	Share	Players
0	Steady Balanced Regulars	15.1%	9,078
1	Dormant / Lapsed Light Accounts	22.3%	13,366
2	High-Value VIP / Whales	18.0%	10,787
3	Casual Low-Stakes Winners	25.2%	15,122
4	High-Spend Net-Loss Gamers	13.6%	8,125
5	Loyal Slow-Converters	5.7%	3,444



- No “noise” points – every profiled player belongs to exactly one segment (unlike HDBSCAN, which would leave 31k unassigned).

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Segment 0 – Steady Balanced Regulars

15.1% of players

The dependable core – steady spenders who give more than they take, on an even keel.

Product view

- Everyday loyalty and tier-progression; they respond to being rewarded for consistency, not to chase-the-bonus mechanics.

Marketing view

- Upsell into the mid-tier and cross-sell between products; sources are balanced (Organic / Referral / Google).

Business view

- Baseline profit engine – healthy margin per deposited dollar (1.04 GGR/£1) with the lowest risk profile; ideal growth pool.

Key numbers (median)

Metric	Value
Deposits	£280
Withdrawals	£50
Bets	1,541
Logins	52
Net position	+£125
GGR	£218
Tenure	101 d
Active	90%

Watch out:

Long-running loyalty is healthy, but 50% are already idle >60 days

Segment 1 – Dormant / Lapsed Light Accounts

22.3% of players

Light-touch, faded-out accounts – they are around, but barely move.

Product view

- Account hygiene first: merge/close duplicate and suspended accounts; archive truly stale profiles instead of paying to keep them.

Marketing view

- Guarded win-back only – they have the 2nd-highest bonus-abuse rate (15.6%), so reactivation offers must carry strict terms.

Business view

- Commercially negligible (0.1% of GGR) and costly through bonus leakage

Key numbers (median)

Metric	Value
Deposits	£20
Withdrawals	£42
Bets	27
Logins	3
Net position	-£12
GGR	£4
Tenure	184 d
Bonus abuse	15.6%

Watch out:

87.7% sit in the NegativeValue profit segment

Segment 2 – High-Value VIP / Whales

18.0% of players

High-Value VIPs – the whales who fund the business (and deserve white-glove care).

Product view

- Dedicated VIP manager, fast withdrawals, personal limits and bespoke journeys – not reload-bonus churn.

Marketing view

- Premium acquisition matters: they arrive via Google/paid search – defend and scale that channel with dedicated creative.

Business view

- Strongest economics in the book – ~90% of total GGR – but the highest formal exit rate and self-exclusion; protect the player to protect the value.

Key numbers (median)

Metric	Value
Deposits	£8,180
Withdrawals	£5,328
Bets	45,351
Logins	413
Net position	+£2,121
GGR	£3,144
Self-excluded	7.0%
RG med/high	11.1%

Watch out:

Their GGR margin is lower (0.38) – they win more back. Priority is retention + RG monitoring, not volume.

Segment 3 – Casual Low-Stakes Winners

25.2% of players

Small-stakes, frequent small wins – the cleanest, cheapest segment to serve.

Product view

- Light-touch engagement: leaderboards, in-account “winning” moments and safe cross-sell – zero deposit-bonus dependence.

Marketing view

- Virtually no bonus abuse (1.3%) – keep them out of bonus churn; low-frequency, high-relevance comms beat volume.

Business view

- Best margin per deposited dollar (1.28 GGR/£1) even at tiny stakes – the margin champion despite small ticket sizes.

Key numbers (median)

Metric	Value
Deposits	£45
Withdrawals	£0
Bets	195
Logins	4
Net position	+£40
GGR	£60
Tenure	208 d
Bonus abuse	1.3%

Watch out:

Biggest blind spot: 76% idle >60 days – win-back / Day3 deposit reminders are the main lever here.

Segment 4 – High-Spend Net-Loss Gamers

13.6% of players

High spend, consistent net losses, and a red-flag behaviour profile – they cost the business money.

Product view

- Restrict incentives: lower or remove bonus eligibility and impose wagering/velocity caps on this profile.

Marketing view

- Do not market them as VIPs; engage with loss-limit nudges and cooling-off prompts rather than offers.

Business view

- The only segment with NEGATIVE GGR (-3.1% of total): they withdraw more than they deposit despite heavy betting – manage, don't grow.

Watch out:

93% sit in NegativeValue profit segment – this is a risk-management cohort, not a growth cohort.

Key numbers (median)

Metric	Value
Deposits	£435
Withdrawals	£670
Bets	1,875
Logins	85
Net position	-£150
GGR	-£7
Win rate	~100%
Bonus abuse	16.9%

Segment 5 – Loyal Slow-Converters

5.7% of players

Loyal and careful – they stay for years but take their time before the first deposit.

Product view

- Fix the onboarding journey: median 74 days from registration to first deposit – a welcome funnel with structured first-deposit offers.

Marketing view

- Google + Facebook driven (38% + 14%) – A/B the sign-up-to-deposit funnel there; highest MFA adoption (3.6%) = cautious, trust first.

Business view

- Longest tenure (369 d) and consistently positive net position – converting them earlier lifts lifetime value materially.

Key numbers (median)

Metric	Value
Deposits	£596
Withdrawals	£248
Bets	3,196
Logins	115
Net position	+£145
GGR	£285
Tenure	369 d
First deposit	74 d

Watch out:

Small cohort by share (5.7%) but the slow-conversion gap is the single most fixable leak in the journey.

Step 6 · Do the clusters actually differ?

Business validation

Yes – and not by a little. The table shows the metrics that matter, per segment.

Metric	C0 Reg.	C1 Dorm.	C2 Whales	C3 Casual	C4 Net-Loss	C5 Slow
Median GGR	£218	£4	£3,144	£60	-£7	£285
Share of total GGR	3.9%	0.1%	89.8%	3.0%	-3.1%	6.4%
GGR per £1 deposit	1.04	0.12	0.38	1.28	-0.02	0.70
Active span (median)	101 d	184 d	331 d	208 d	179 d	369 d
Formal churn/Self-excl.	7.8%	6.0%	16.7%	12.2%	8.9%	10.2%
Idle >60 days	50%	72%	63%	76%	60%	65%
Bonus dependency	5.4%	15.6%	4.7%	1.3%	16.9%	5.9%
Tenure (median)	101 d	184 d	331 d	208 d	179 d	369 d
Logins (median)	52	3	413	4	85	115

- A ~1,000× gap in GGR between segments (C2 £3,144 vs C1 £4) – commercial difference is extreme.
- Type of churn differs too: Whales leave formally (self-exclusion/timeout), Casual players just fade away.

What can be done differently (recommendations)

Actions

#	Action	Target segment	Why first
1	Protect & safeguard – VIP manager, RG monitoring, self-limit offers	C2 Whales	~90% of GGR at risk; exit 16.7%
2	Restrict bonuses + loss-limit nudges	C4 Net-Loss	Only negative-GGR cohort; highest abuse
3	Clean-up + guard-railed win-back	C1 Dormant	Lowest effort, stops abuse leakage
4	Tier-progression & cross-sell program	C0 Regulars	Healthy core, cheapest to grow
5	Optimise onboarding / first-deposit funnel	C5 Slow-Converters	74-day first-deposit gap
6	Light-touch engagement mechanics	C3 Casual	Large base, margin-first growth

- Cross-cutting: segment-specific incentives only (no one-size bonuses), cluster labels into the CRM/CDP, governed A/B tests before full rollout, RG thresholds tuned per segment spend-band.

Caveats – what this model is NOT claiming

Limitations

- 40,078 players (40%) had too much missing data to profile – they have no cluster label. That means “not assessed for modelling”, NOT “low value”.
- Hierarchical and DBSCAN were fitted on subsamples (6k / 15k players) for computing reasons – their segment boundaries depend on that sample.
- Many risk/abuse booleans were never recorded (e.g. bonus abuse recorded for only ~6% of players) – flagged as “unknown”, never assumed clean.
- With 59,922 rows, significance tests are trivially “significant” – we therefore relied on effect sizes and stability, not just p-values.
- Recommendations are hypotheses. Validate each with segment-targeted A/B tests before full rollout.

Appendix – technical details

For the data team

Stability of the deployed models (mean over 10 pairwise runs)

Model	Perturbation	ARI mean	ARI ±	NMI mean	NMI ±
K-Means (k=6)	random seeds, full data	0.996	0.002	0.993	0.003
Gaussian Mixture (k=3)	random seeds, full data	0.804	0.240	0.834	0.204
Hierarchical (Ward)	5 draws, n=6,000	0.579	0.119	0.595	0.077
DBSCAN	5 draws, n=15,000	0.855	0.015	0.710	0.023
HDBSCAN	5 draws, n=30,000	0.640	0.201	0.698	0.101

Distances used by each algorithm (see distance_measures.md)

- K-Means – Euclidean (squared)
- GMM – Mahalanobis (per-cluster covariance)
- Hierarchical – Ward (ESS increase)
- DBSCAN – core-distance reachability
- HDBSCAN – mutual reachability distance

Data flow and files

- raw CSV (100,000×77) → EDA → 43 derived features → 10 features, log/z-scored → 59,922 complete-case matrix (feature_matrix_11_complete.csv).
- All 5 models fit on the same matrix; labels per model in *_assignments.csv; cluster-labelled raw data in master_dataset_clusters.csv.
- Viz: PCA scatters (2D 58.5% / 3D 70.2% explained variance, viz-only), 3D rotating GIFs, elbow/silhouette/dendrogram/AIC-BIC, model & stability comparison charts.